Zigang Zhu

Ph.D. Candidate in Mathematics · Algebraic Geometry

Email: zhzg0313@tju.edu.cn
Homepage: zigangzhu.github.io



Education

Tianjin University, Center for Applied Mathematics, *Ph.D. in Mathematics* Sep. 2023–Jun. 2027 (expected)
- Advisor: Professor Xun Yu

Tianjin University, Center for Applied Mathematics, *M.S. in Mathematics* Sep. 2020–Jul. 2023
- Advisor: Professor Xun Yu
- Master's thesis: *Automorphism Groups of Smooth Cubic Fivefolds*

The University of Manchester, *B.Sc. in Mathematics, First-Class Honours (joint program)* Sep. 2018–Jun. 2020

Tianjin University, *B.S. in Mathematics and Applied Mathematics* Sep. 2016–Jun. 2020

Research Interests

Automorphism groups of smooth hypersurfaces; rationality and unirationality of algebraic varieties and their quotients

Publications

1. *Automorphism groups of cubic fivefolds and fourfolds*.
   Song Yang, Xun Yu, and **Zigang Zhu**. *Journal of the London Mathematical Society* (2) **110** (2024), no. 4, e12997, 35 pp. arXiv:2308.07186.
2. *On automorphism groups of smooth hypersurfaces*.
   Song Yang, Xun Yu, and **Zigang Zhu**. *Journal of Algebraic Geometry* **34** (2025), 579–611. arXiv:2405.09505.
3. *Nonrational varieties with unirational parametrizations of coprime degrees*.
   Song Yang, Xun Yu, and **Zigang Zhu**. Preprint, 2025. arXiv:2508.03623.
4. *A smooth quintic surface with 85 lines and Picard number 43*.
   Xun Yu and **Zigang Zhu**. Preprint, 2026. arXiv:2609.26862.
5. *Automorphism groups of smooth quartic threefolds*.
   Zigang Zhu. Preprint, 2026. arXiv:2610.39627.

Talks

- *Nonrational varieties with unirational parametrizations of coprime degrees*. Wuhan University, Wuhan, China, Apr. 2026.
- *Computer algebra in the study of automorphism groups of smooth hypersurfaces*. Multi-zeta Values and Iterated Integrals Workshop, Tsinghua University, Beijing, China, Nov. 2023.

Teaching Experience

Tianjin University, Teaching Assistant 2020–2024
Topics in Advanced Algebra (2020); Abstract Algebra (2021); Advanced Mathematics (2022, 2023); Calculus I (2024)
Led tutorials, graded assignments, and held office hours.

Conferences and Summer Schools

- National Conference on Algebraic Geometry, Shandong University, 2024
- Group Actions in Algebraic Geometry, Morningside Center of Mathematics, 2025
- International Summer School on Algebraic Geometry, Fudan University, 2025
- The 10th International Congress of Chinese Mathematicians (ICCM 2025), Shanghai, China, Jan. 2026
- The 5th National Conference on Algebraic Geometry, Peking University, Beijing, China, Jul. 2026

Honors and Awards

Outstanding Master's Thesis of Tianjin 2025
National Scholarship for Doctoral Students 2025
Tianjin University Academic Scholarships 2021–2026

Skills

Mathematical software: SageMath, Mathematica, GAP, and Macaulay2
Languages: Mandarin Chinese (native); English (proficient)